

PRINCESS CROAK'S PALACE · PRINCESS CROAK · EVERY UNIT, EVERY DIFFICULTY

What every gnome, fairy, wasp, frog and Princess Croak herself does to your men on EASY, MEDIUM, HARD and MEGA, with every body's HP and head-count · read out of the live game v32.36 on Nov 14, 2026 03:08 EST · every damage number checked on the game's own machines (Chrome 309/309 · WebKit 309/309) · for Raziya

The one thing to know: since v32.27 **her harp shot kills**: a shot that lands fells any man on foot, whatever his level, and takes half a heavy's bar (the bazooka, the flame and the commando, every machine: two shots; on MEGA a heavy below level 70 falls to one); **her tongue's jump-back crit fells every man she licks** (every 14 · 11 · 9 · 7 s, the nearest man within 380 px); and **her court comes twice** (98 · 118 · 272 · 344 bodies in the waves). Sarge and the heroes take her shot as **bar hits** (2, MEGA 3), and so does every hit of her court: the guard's jab (1), the knight's jab (2), the slinger's glob (1), the wasp's sting (1, and the man is held 2 s), the hornet's sting (3, and he is bowled over), the frog's splat and tongue (1 each), the sage's fire (1 a beat, 10 beats a ring) and his fireball (2 within 60 px), each landing at 50 % on your men (THE DAMPER). **No hit is an instant KO** (since v29.70): the gnome guard's RAM and her FACE SHOT take 49.5 HP (MEGA 58.5) landing whole, and no single hit fells a man other than a rifleman from a full bar (he stands on 15 % of it). **Her LIGHT ARROWS push** every man they cross 220 px west and kill nobody; the sage's toad curse takes Sarge out of the fight for 5 s. Difficulty sets her bar by the designers' time-to-kill chart on EASY · MEDIUM · HARD and the player-damage-threshold pass on MEGA (**2,706 · 3,608 · 1,804 · 7,560** with one commander), her lives (1 · 1 · 3 · 3: one full bar on EASY and MEDIUM, three full bars on HARD and MEGA, so 2,706 · 3,608 · 5,412 · 22,680 in all), her cadence, the court's size and HP, and brings the knights, the sages and the hornets.

THE BOSS LAB fights the palace at **LEVEL 70 · 77 · 83 · 90** (EASY · MEDIUM · HARD · MEGA): Sarge with the saber, the scout jeep, the dogfighter, the tank, the bomber and seven men at that rank (rifleman, M. gunner, bazooka, mortar, medic, flame, sweeper). The radio man, grenadier, commando and spy are a tap away in THE PLATOON. Sarge wears the two rank-5 gifts (telekinesis on the saber; the riot shield answers only with the shotgun in hand). Every "level" number on this page is that army.

What changed recently (the palace's own cards)

live since	when	card	what it does in this fight
v32.27	Nov 13	**CARD-PCH · PRINCESS CROAK HITS HARD**	Wyatt's notes from his co-op test with Isaac. HER AUTO ATTACK KILLS: a harp shot that lands takes at least a man's whole bar on foot and half a heavy's (the bazooka, the flame and the commando, every machine; on MEGA a heavy below level 70: all of it). Was → now at the lab's level on EASY: a rifleman 8 → 1 shot, a medic 7 → 1, a flame 8 → 2, a commando 9 → 2 (MEGA: a rifleman 6 → 1, a bazooka 6 → 2); Sarge and the heroes as before. THE JUMP-BACK CRIT: the man her tongue lands at her feet takes her hop back and one shot that always crits, and falls (measured 0.31–0.32 s after the landing); the yoink's own bar hit is gone. TWICE THE COURT: every body on her side ×2: the lab's sheet 49 · 59 · 136 · 172 → 98 · 118 · 272 · 344 bodies (E · M · H · G); THE BUBBLE COURT HARD 3 + 3 → 6 + 6, MEGA 4 + 4 → 8 + 8.
v32.25	Nov 13	**CARD-PCS · ONE FIGHT FOR THE PARTY**	In co-op her bar is the referee's on every phone: her fall and her retreats no longer come apart between two phones, and a partner who leaves never freezes the one who stayed. No number on this page moves.
v31.45	Nov 11	**CARD-PYK · HER TONGUE YOINK**	Every 14 · 11 · 9 · 7 s she snatches the nearest man within 380 px with a clear line out of your line and reels him to her feet (one bar hit and a 1 s daze, never the felling one, until v32.27's jump-back crit); 5 % of her bar landed during the reel snaps the tongue; a Dogfighter is pulled down when no man is in reach.
v31.40	Nov 11	**CARD-PKB · HER ARROWS PUSH, THE COURT CASTS**	Her light arrows push every man they cross 220 px west instead of the KO (a machine 0.3 of it, never into the water; a column bumps, the pool is a fall); EASY gets the arrow with its half-bar release; the wave that brings the arrow spouts its ground court at the two fountains as the bolts fly; one ground group a wave arrives by THE TELEPORT RING near your line (MEDIUM up; EASY from its 3rd return); the sage hurls a FIREBALL (2 bar hits within 60 px and a fire patch, every 9 s); the bazooka's rockets take ×2.4 off the gnome knight.
v31.39	Nov 10	**CARD-PFL · THE PALACE FLOWS**	Her waves come on her bar: a wave holds while her court stands, then comes at her bar's mark or after the idle clock (56 · 48 · 42 · 42 s, the old gate clock counted with her court down only); THE COURT RETURNS come on her bar below the last gate (EASY at 25 · 20 · 15 · 10 · 5 %); a wave cleared pays 120 · 150 · 180 · 200 dog tags; the Boss Lab's TRY AGAIN keeps the lab.

live since	when	card	what it does in this fight
v31.38	Nov 10	**CARD-PTP · THE PILLARS STAND TALL**	No man walks over a column's or a fountain's top any more and a man behind one is drawn behind it; their footprints block the walk, the jeep, the ram and the push.
v31.00	Nov 6	**CARD-RZB · RAZIYA'S BOSS NOTES (every boss fight)**	No single hit fells a man other than a rifleman from a full bar: he stands on 15 % of it and the next one fells him; one blow reaches at most 3 · 4 · 4 · 5 men (E · M · H · G).
v29.70	Nov 3	**CARD-BDC2 · NO INSTANT KO (Isaac's pass)**	A blow that KO'd outright (the guard's ram, her face shot, her light arrows until v31.40) takes 0.9 × the M. gunner's sheet HP at the chart's rank for the palace's level (140, rank 80; MEGA 180, rank 100) instead: 49.5 HP (MEGA 58.5), landing whole. Was → now on a level-70 man, the ram: a rifleman KO → KO, a flame KO → 3 rams, a pistol Sarge KO → 2 (with v31.00's floor). The heavy class (the commando, the flame, the bazooka) carries at least ×2 the M. gunner's HP (at the lab's levels their own ladder is higher already).
v29.41	Nov 2	**CARD-DHP · THE BONUS DOUBLED FOR TWO, THE SHOTGUN'S HEALTH**	The bazooka's and the commando's tenth-rank HP bonus doubled (5 → 10 HP a rung); shotgun Sarge +20 HP per ten shotgun ranks while the shotgun is in his hands. At level 70 · 77 · 83 · 90 (EASY · MEDIUM · HARD · MEGA), was → now: the bazooka 66 → 101 · 66 → 101 · 75 → 115 · 83 → 128 HP, the commando 75 → 110 · 75 → 110 · 84 → 124 · 93 → 138, the shotgun Sarge 40 → 180 · 41 → 181 · 44 → 204 · 47 → 227. Every palace hit counts in bar hits, so each bar hit on them is worth more HP; their counts to KO stand.
v29.40	Nov 2	**CARD-FLT · THE FLAME'S TENTH-RANK HP TRIPLED THROUGH LEVEL 50**	The flame's tenth-rank HP rungs through level 50 pay 18 instead of 6 (the first 5; the rungs past level 50 keep 6): the flame 80 → 140 · 80 → 140 · 90 → 150 · 100 → 160 HP at level 70 · 77 · 83 · 90. His bar hits are unchanged, so her harp KOs him on the same shot.
v29.37	Nov 2	**CARD-RXT · THE DESIGNERS' TIME-TO-KILL CHART**	Her bar for one commander on EASY · MEDIUM · HARD is the chart's number; the dial that multiplied her 160 HP before v29.30 (×2 · ×2.4 · ×6 · ×9) no longer does (the player-damage-threshold pass had moved these three bars first at v29.30; the chart wrote its own numbers over them at v29.37). Was → now: 320 → 2,706 on EASY, 384 → 3,608 on MEDIUM, 960 → 1,804 on HARD (each of her three bars there: 2,880 → 5,412 in all). A party's bar follows at solo × (1 + 0.9 a commander past the first). MEGA stays the player-damage-threshold pass's.
v29.30	Nov 2	**CARD-PDT · THE PLAYER DAMAGE THRESHOLD**	The bosses' bars for one commander set from the average player's damage threshold at the fight's commander level: her MEGA bar 1,440 → 7,560 (each of her three bars there: 4,320 → 22,680 in all); the dial ×9 over her 160 HP no longer sets it. Its EASY · MEDIUM · HARD cells stood until the time-to-kill chart (v29.37) wrote those three again.
v29.17	Nov 1	**CARD-DMP · THE DAMPER**	Every enemy blow on your men lands at 50 % in the campaign and the boss fights, the Boss Lab included (the battleground 70 %). A palace bar hit now lands 50 % × (90 % + 10 % × bar ÷ HP) of its weight, so a man stands about twice the blows: a level-70 rifleman falls to her 8th harp shot on EASY (the 4th before) and to a guard's 16th jab (the 8th). The ram, the light arrows and the face shot are KOs and land as shipped.
v28.93	Oct 31	**CARD-KCF · THE ELUSIVE CROAK**	The King Croak hero carries more HP: at level 70 he has 337 HP and a bar of 297 (the heroes' table).
v28.74	Oct 31	**CARD-HP3 · THE HEAVIES**	+3 HP every tenth rank for the bazooka, grenadier, flame and commando. At level 70 before this afternoon's cards (the v29.25 read): bazooka 66, grenadier 61, flame 80, commando 75 HP; today 101 · 61 · 140 · 110 (the rows above). Every palace hit counts in bar hits, so more HP means each bar hit is worth more HP, not fewer hits.
v28.68	Oct 30	**CARD-BRB · THE BOMBER**	Your bomber flies in every boss fight now, the palace included.
v28.65	Oct 30	**CARD-LRB · ONE HP NUMBER**	Your man's HP is the number on his unit sheet. A palace hit still counts in bar hits, but lands a little lighter on tough men: 90 % + 10 % × bar ÷ HP of its weight (a level-70 rifleman takes 0.92 of 2 from her harp, a flame 0.91).
v28.46	Oct 30	**CARD-BLP · THE BOSS LAB'S STOCK ARMY**	The lab fights the palace with its own army at LEVEL 70 · 77 · 83 · 90 (EASY · MEDIUM · HARD · MEGA): the one this page reads.
card 1213	—	**CARD-LA2 · THE KNIGHT'S MIRROR**	A knight's shield met from the front by a Sarge's own light arrow (the sniper mod won on HARD) throws it back at the shooter.
card 1027	—	**CARD-ML-A3 · PRINCESS CROAK, THE SIXTH HERO**	Beat her on MEGA and she joins as a hero (her HP row sits among the heroes above).

live since	when	card	what it does in this fight
card 1023	Oct 23	**CARD-PCS · PALACE SPACING**	Laura's spacing pass on her palace picture (the look; no fight numbers).
10/10	Oct 10	**CARD-PC2 · THE BIGGER COURT**	Wyatt: "give her more troops coming in" and "There should never be two Princess Croaks out at once": HARD and MEGA field more guards, slingers, wasps, knights, frogs; two hornets on HARD, three on MEGA; three sages at MEGA's open; THE COURT RETURNS ×8 on HARD and MEGA; one princess at every seat count.
card 1014	—	**THE PRESSURE · THE PALACE TUNE**	Her harp shot per tier (the cadence 2.4 · 2.0 · 1.8 · 2.0 s, MEGA 3 bar hits), the gate clocks, the knights at 75 %, the guard turning out at 50 % and 30 %, THE COURT RETURNS on the clock, the court's HP cells.
CARD-P3	—	**THE RETREAT · THE BUBBLE COURT**	HARD and MEGA: three bars; at a zero with bars left she flies back to her stage and calls her fairies and guards; the bar clock; the big guards charge at ×2.2.
card 1032	Sep 14–15	**CARD-PM-A · THE THRONE ROOM**	Wyatt: "Princess croaks map also needs to be redesigned to feel more epic". Six marble columns as cover, two frog fountains, the water stairs as a river with two bridges, the pool, her throne and harp behind her top step; MEGA moonlit.
card 0947e	Sep 11	**THE PALACE RAID**	Wyatt: "at half HP point, is going to begin to charge her light arrow… three large light arrows… if a commando has his reflect, you may reflect it right back at her… built-in auto-kill mechanics for any Sarge that comes too close". The fight as designed.
—	—	**THE FIRST READ**	The first read (Oct 31, v28.88) was built on Maker 1 from a shallow clone of the Big Mac's main 04d094bb69, with no git log there: the rows from v28.74 down were read off the client's own comments and the other kits' records, with a version where one is known. This re-read (THE DAMPER) is the Big Mac's, on its own client.

How a hit is counted

Your men carry **HP**: the number on their unit sheet (a level-70 flame has 140, a rifleman 33). Their health bar is also cut into a few **bar hits** (7 for most men at level 70–90, 6 for the medic and spy, 8 for the commando, 12 for the saber Sarge, 23 for the Claymore). Three kinds of hit in this fight:

- **Bar hits** (everything the court does, and her harp): a hit of N bar hits lands 90 % + 10 % × bar ÷ HP of N since v28.65, and since v29.17 50 % of that (THE DAMPER, below), so a level-70 rifleman takes 0.92 of his 7 from her harp (4.3 HP of 33) and 0.46 from a guard's jab. Bar hits mend after 4 s without a hit. A hero's bar IS his HP pool (a bar hit lands on it with no share, at the damper's 50 %).
- **THE DAMPER** (since v29.17): every blow an enemy lands on your men in the campaign and the boss fights, the Boss Lab included, lands at **50 %** of its size; it rides every bar hit on this page, the heroes' too, and every count and level here is counted with it. A KO hit is not dampened: it lands as shipped.
- **No instant KO** (since v29.70, Isaac's pass): the guard's ram and her face shot take 49.5 HP (MEGA 58.5) off a man's sheet, landing whole (the damper not taken), as bar hits = that HP × bar ÷ HP; and since v31.00 no single hit fells a man other than a rifleman from a full bar: the blow that would empty it leaves him on 15 %, the next one fells him. So a level-70 rifleman (33 HP) falls to one ram, a flame (140 HP) to 3 (measured: the first took 2.475 of his 7 bar hits), a pistol Sarge to two (measured: the first left him on 1.65 of 11); the saber turns the ram and the face shot about half the time (measured 14/15 · 12/12 · 11/15 · 12/11 of 16 rams turned per tier, Chrome/WebKit), THE RIOT SHIELD (5 a battle, the shotgun) and the Claymore's counted block (1 a battle: THE DUELIST, every Claymore's default spec) eat them whole (measured: the shotgun Sarge fell to ram 9 · 9 · 10 · 9, the Claymore to ram 3 · 3 · 3 · 3, E · M · H · G); the pet shield, the blue wall and the teal turn the ram; the deflect dome turns her face shot. A jeep or the tank on the guard's spear is still destroyed. **Two hits still fell outright**: her harp shot on a man on foot (a heavy's second) by her auto attack's law, and her tongue's jump-back crit.
- **No hit**: the fairy mends her court and bubbles a guard BIG; the bubble itself pops on a green round; the sage's curse turns Sarge into a toad for 5 s (no fire, no swing).

Saves still roll on every bar hit: a jetting, dropping or dragged man cannot be touched, the pet shield, the deflect dome, **the saber Sarge's block turns a bar hit 50 % of the time** (measured: 27 of 61, 66 of 125, 10 of 26, 8 of 20 of her shots on the four tiers), **the shotgun Sarge's riot shield eats the first 5 hits whole** (measured), and the Claymore's counted block eats 1 (measured; THE DUELIST's one block is every Claymore's default spec, the table's 3 only with FEAR or WALL picked). A shot of hers that a save turns takes nothing, her auto attack's law included; the same saves answer the priced ram and face shot (the bite runs the bar hits' door). A charged commando catches a light arrow and sends it back at her.

The strikes · what a hit does by difficulty

hit	EASY	MEDIUM	HARD	MEGA	who it hits	since
HER HARP SHOT (her auto attack)	**a man on foot: KO · a heavy: 2**	**a man on foot: KO · a heavy: 2**	**a man on foot: KO · a heavy: 2**	**a man on foot: KO · a heavy: 2 (below level 70: KO)**	the nearest man within 400 px, every 2.4 · 2 · 1.8 · 2 s: a shot that lands takes his whole bar (a heavy: the bazooka, the flame and the commando, every machine, half); Sarge and the heroes take 2 · 2 · 2 · 3 bar hits a shot	HER AUTO ATTACK KILLS: v32.27 (CARD-PCH); the cadence and MEGA's third bar hit: card 1014
HER TONGUE YOINK · the jump-back crit	**KO**	**KO**	**KO**	**KO**	the nearest man within 380 px with a clear line, every 14 · 11 · 9 · 7 s: reeled to her feet, her hop back and one shot that always crits (never Sarge, a hero or a machine)	the yoink v31.45 (CARD-PYK); the crit v32.27 (CARD-PCH)

hit	EASY	MEDIUM	HARD	MEGA	who it hits	since
HER FACE SHOT · HP	**49.5**	**49.5**	**49.5**	**58.5**	a Sarge within 84 px: the hop back and the shot, landing whole; his full bar's first bite leaves him on 15 %; the saber turns it about half the time, THE RIOT SHIELD's 5 and the Claymore's counted block eat it	no instant KO v29.70 (CARD-BDC2); the floor v31.00 (CARD-RZB)
HER LIGHT ARROWS · a push	**220 px**	**220 px**	**220 px**	**220 px ×2**	every man a bolt's line crosses is pushed west (a machine 0.3 of it); a column or a fountain stops him with a bump, the pool or the water is a fall; the light kills nobody; a charged commando, the blue wall or the teal catches one	the push and EASY's arrow v31.40 (CARD-PKB); MEGA's second volley: card 1014

hit	EASY	MEDIUM	HARD	MEGA	who it hits	since
THE GNOME'S RAM · HP	**49.5**	**49.5**	**49.5**	**58.5**	a man ahead within 300 px, landing whole: a rifleman falls, any other man stands his first from a full bar on 15 % of it; a jeep or the tank is destroyed on the spear; the saber turns it about half the time, THE RIOT SHIELD's 5 and the Claymore's counted block eat it, the pet shield, the blue wall and the teal turn it	no instant KO v29.70 (CARD-BDC2); the floor v31.00 (CARD-RZB)
THE GNOME'S JAB · bar hits	**1**	**1**	**1**	**1**	a man within 28 px	unchanged
THE KNIGHT'S JAB · bar hits	**—**	**2**	**2**	**2**	a man within 28 px	unchanged
THE SLINGER'S GLOB · bar hits	**1**	**1**	**1**	**1**	a man within 280 px	unchanged
THE SAGE'S FIRE · bar hits a beat	**—**	**1**	**1**	**1**	your three rearmost men, 10 beats over 5 s	unchanged

hit	EASY	MEDIUM	HARD	MEGA	who it hits	since
THE SAGE'S FIREBALL · bar hits	—	2	2	2	every man within 60 px of where it lands, at the densest clump within 420 px, every 9 s; a fire patch there	v31.40 (CARD-PKB)
THE SAGE'S TOAD CURSE	—	5 s a toad	5 s a toad	5 s a toad	a Sarge within 200 px: no fire, no swing, hops at ×0.6	unchanged
THE WASP'S STING · bar hits	1	1	1	1	every man a dive crosses; held 2 s	unchanged
THE HORNET'S STING · bar hits	—	—	3	3	every man a dive crosses; bowled over	two hornets on HARD, three on MEGA: CARD-PC2
THE FROG'S SPLAT · TONGUE · bar hits	1 · 1	1 · 1	1 · 1	1 · 1	a man within 150 px (the hop) · 56 px (the tongue)	unchanged
THE FAIRY'S BUBBLE	BIG gnome: ×4 HP	BIG gnome: ×4 HP	BIG gnome: ×4 HP	BIG gnome: ×4 HP	her own court, never your men	unchanged

The blows as the game ships them. On your men every one lands at 50 % (THE DAMPER, since v29.17: the campaign and the boss fights, the Boss Lab included), times the share; the ram's and the face shot's bites land whole (no instant KO since v29.70).

Who survives · the lab's level

How many of that hit in a row (inside the 4 s mend) KO a fresh man at the lab's level for that tier (EASY 70 · MEDIUM 77 · HARD 83 · MEGA 90), each blow landing at the damper's 50 %. **KO** = one hit. Hover a cell for the numbers.

KO one hit · 2 two · 3 three · 4 four · 5+ five or more

· a dot in the corner = the count measured on the game's own machines in this read (both engines agree) · +½ = the saber turns about half (her harp shot, the ram, the face shot) · +5 = the riot shield eats 5 first · +1 on the Claymore = his counted block eats 1 first · "·" = not in this fight / only a Sarge

man	HP E · M · H · G	bar hits	HER HARP SHOT E	HER HARP SHOT M	HER HARP SHOT H	HER HARP SHOT G	GNOME'S JAB E	GNOME'S JAB M	GNOME'S JAB H	GNOME'S JAB G	KNIGHT'S JAB E	KNIGHT'S JAB M	KNIGHT'S JAB H	KNIGHT'S JAB G	SLINGER'S GLOB E	SLINGER'S GLOB M	SLINGER'S GLOB H	SLINGER'S GLOB G	WASP'S STING E	WASP'S STING M	WASP'S STING H	WASP'S STING G	HORNET'S STING E	HORNET'S STING M	HORNET'S STING H	HORNET'S STING G	FROG'S SPLAT · TONGUE E	FROG'S SPLAT · TONGUE M	FROG'S SPLAT · TONGUE H	FROG'S SPLAT · TONGUE G	SAGE'S FIRE (beats) E	SAGE'S FIRE (beats) M	SAGE'S FIRE (beats) H	SAGE'S FIRE (beats) G	SAGE'S FIREBALL E	SAGE'S FIREBALL M	SAGE'S FIREBALL H	SAGE'S FIREBALL G	GNOME RAM E	GNOME RAM M	GNOME RAM H	GNOME RAM G	FACE SHOT E	FACE SHOT M	FACE SHOT H	FACE SHOT G	TONGUE YOINK E	TONGUE YOINK M	TONGUE YOINK H	TONGUE YOINK G
THE LAB'S STOCK ARMY (LEVEL 70 · 77 · 83 · 90)																																																					
RIFLEMAN	33 · 33 · 37 · 41	7	KO	KO	KO	KO	16	16	16	16	·	8	8	8	16	16	16	16	16	16	16	16	·	·	6	6	16	16	16	16	·	16	16	16	·	8	8	8	KO	KO	KO	KO	·	·	·	·	KO	KO	KO	KO
M. GUNNER	50 · 50 · 55 · 60	7	KO	KO	KO	KO	16	16	16	16	·	8	8	8	16	16	16	16	16	16	16	16	·	·	6	6	16	16	16	16	·	16	16	16	·	8	8	8	2	2	2	2	·	·	·	·	KO	KO	KO	KO
BAZOOKA	101 · 101 · 115 · 128	7	2	2	2	2	16	16	16	16	·	8	8	8	16	16	16	16	16	16	16	16	·	·	6	6	16	16	16	16	·	16	16	16	·	8	8	8	3	3	3	3	·	·	·	·	KO	KO	KO	KO
MORTAR	30 · 30 · 33 · 37	7	KO	KO	KO	KO	16	16	16	16	·	8	8	8	16	16	16	16	16	16	16	16	·	·	6	6	16	16	16	16	·	16	16	16	·	8	8	8	2	2	2	2	·	·	·	·	KO	KO	KO	KO
MEDIC	29 · 29 · 33 · 36	6	KO	KO	KO	KO	14	14	14	14	·	7	7	7	14	14	14	14	14	14	14	14	·	·	5	5	14	14	14	14	·	14	14	14	·	7	7	7	2	2	2	2	·	·	·	·	KO	KO	KO	KO
FLAME	140 · 140 · 150 · 160	7	2	2	2	2	16	16	16	16	·	8	8	8	16	16	16	16	16	16	16	16	·	·	6	6	16	16	16	16	·	16	16	16	·	8	8	8	3	3	4	3	·	·	·	·	KO	KO	KO	KO
SWEEPER	31 · 31 · 35 · 38	7	KO	KO	KO	KO	16	16	16	16	·	8	8	8	16	16	16	16	16	16	16	16	·	·	6	6	16	16	16	16	·	16	16	16	·	8	8	8	2	2	2	2	·	·	·	·	KO	KO	KO	KO
SARGE · SABER	40 · 41 · 44 · 47	12	13+½	13+½	13+½	9+½	26	26	26	26	·	13	13	13	26	26	26	26	26	26	26	26	·	·	9	9	26	26	26	26	·	26	26	26	·	13	13	13	2+½	2+½	2+½	2+½	2+½	2+½	2+½	2+½	·	·	·	·
A TAP AWAY (THE PLATOON, SARGE'S OTHER IRONS)																																																			
RADIO MAN	31 · 31 · 35 · 38	7	KO	KO	KO	KO	16	16	16	16	·	8	8	8	16	16	16	16	16	16	16	16	·	·	6	6	16	16	16	16	·	16	16	16	·	8	8	8	2	2	2	2	·	·	·	·	KO	KO	KO	KO
GRENADIER	61 · 61 · 68 · 76	7	KO	KO	KO	KO	16	16	16	16	·	8	8	8	16	16	16	16	16	16	16	16	·	·	6	6	16	16	16	16	·	16	16	16	·	8	8	8	2	2	2	2	·	·	·	·	KO	KO	KO	KO
COMMANDO	110 · 110 · 124 · 138	8	2	2	2	2	18	18	18	18	·	9	9	9	18	18	18	18	18	18	18	18	·	·	6	6	18	18	18	18	·	18	18	18	·	9	9	9	3	3	3	3	·	·	·	·	KO	KO	KO	KO
SPY	36 · 36 · 41 · 45	6	KO	KO	KO	KO	14	14	14	14	·	7	7	7	14	14	14	14	14	14	14	14	·	·	5	5	14	14	14	14	·	14	14	14	·	7	7	7	2	2	2	2	·	·	·	·	KO	KO	KO	KO
SARGE · PISTOL	41 · 41 · 44 · 47	11	12	12	12	8	24	24	24	24	·	12	12	12	24	24	24	24	24	24	24	24	·	·	8	8	24	24	24	24	·	24	24	24	·	12	12	12	2	2	2	2	2	2	2	2	·	·	·	·
SARGE · SHOTGUN	180 · 181 · 204 · 227	8	9+5	9+5	9+5	6+5	18	18	18	18	·	9	9	9	18	18	18	18	18	18	18	18	·	·	6	6	18	18	18	18	·	18	18	18	·	9	9	9	4+5	4+5	5+5	4+5	4+5	4+5	5+5	4+5	·	·	·	·
SARGE · SNIPER	31 · 31 · 34 · 37	8	9	9	9	6	18	18	18	18	·	9	9	9	18	18	18	18	18	18	18	18	·	·	6	6	18	18	18	18	·	18	18	18	·	9	9	9	2	2	2	2	2	2	2	2	·	·	·	·
SARGE · CLAYMORE	71 · 73 · 73 · 76	23	25+1	25+1	25+1	17+1	50	50	50	50	·	25	25	25	50	50	50	50	50	50	50	50	·	·	17	17	50	50	50	50	·	50	50	50	·	25	25	25	2+1	2+1	2+1	2+1	2+1	2+1	2+1	2+1	·	·	·	·

The bar is 7 for most men from level 70 to 90, so the counts hold across the tiers; HP is what each bar hit is worth. HER HARP SHOT: every man and iron measured on every tier, each to his KO (the dots): a man on foot by her auto attack's law (one shot that lands), a heavy two; Sarge's irons by the arithmetic. The court's hits were measured on a rifleman (jab, glob, sting, frog, fire), a flame (knight, hornet, fire) and a grenadier (fire): each hit's weight on every tier, and the count to the KO (the dot) for the guard's jab, the slinger's glob and the knight's jab; the rest of the column is the same arithmetic. SAGE'S FIRE counts beats in a row; a ring holds 10 (5 s), so a count over 10 means he walks out of the first ring: measured on MEDIUM (level 77), the rifleman with 4.61 of his 7, the flame with 4.53 of his 7, the grenadier with 4.56 of his 7; the 4 s mend gave 1 back before the next ring (12 s on), which KO'd each on its 8th beat (18 beats in all). GNOME RAM and FACE SHOT: the priced bite on every tier (measured on a rifleman, a flame and Sarge's saber, pistol, shotgun and Claymore; the face shot on all four irons; a dot where both engines measured the count to the KO). TONGUE YOINK: measured on a rifleman, a bazooka and a commando on every tier (the dots); every other man by the same crit. SAGE'S FIREBALL: its weight measured on a clump of three men (a fourth 130 px from their middle untouched); the counts are the arithmetic. SARGE · SABER's harp cells carry no dot: 13 · 13 · 13 · 9 is the count if every landed shot came inside the 4 s mend, but his block spaces them and his bar mends between, so the machines measured 34/18 · 59/66 · 16/29 · 12/26 landed shots to his KO (E · M · H · G, Chrome/WebKit, after 147/67 · 252/274 · 47/136 · 40/141 s: at the damper's half his mend nearly keeps pace with her).

The heroes at the lab's level (one in command), measured

hero	HP (E · M · H · G)	bar hits	her harp shot	shots to KO	measured: shots landed in 60 s (Chrome/WebKit where they differ)	her face shot	her tongue
KING YETI	**258 · 259 · 296 · 333**	246 · 246 · 302 · 358	1 · 1 · 1 · 1.5 of his bar a shot	246 · 246 · 302 · 239	24 · 29 · 32 · 29	never: not a Sarge	never: not on her tongue (it licks only a man)
KING REX	**329 · 331 · 351 · 359**	257 · 257 · 283 · 289	1 · 1 · 1 · 1.5 of his bar a shot	257 · 257 · 283 · 193	25 · 29 · 33 · 29	never: not a Sarge	never: not on her tongue (it licks only a man)
KING CROAK	**337 · 339 · 346 · 353**	297 · 297 · 303 · 309	1 · 1 · 1 · 1.5 of his bar a shot	297 · 297 · 303 · 206	25 · 29 · 33 · 29	never: not a Sarge	never: not on her tongue (it licks only a man)
JANE	**126 · 128 · 135 · 142**	68 · 68 · 74 · 80	1 · 1 · 1 · 1.5 of his bar a shot	68 · 68 · 74 · 54	25 · 29 · 33 · 29	never: not a Sarge	never: not on her tongue (it licks only a man)
EMPEROR BEAR	**338 · 341 · 361 · 369**	257 · 257 · 283 · 289	1 · 1 · 1 · 1.5 of his bar a shot	257 · 257 · 283 · 193	21 · 25 · 29 · 21	never: not a Sarge	never: not on her tongue (it licks only a man)
PRINCESS CROAK	**97 · 98 · 104 · 111**	77 · 77 · 83 · 89	1 · 1 · 1 · 1.5 of his bar a shot	77 · 77 · 83 · 60	23 · 27 · 31 · 27	never: not a Sarge (measured at 60 px)	never: not on her tongue (it licks only a man)

A hero's bar is his HP pool: her harp takes 1 (MEGA 1.5) of it a shot: her 2 (MEGA 3) at the damper's 50 %, with no share (measured on every tier). No hero fell inside 60 s on any tier (the damper halves her shot on the pool too). A hero walks, so the count landed in 60 s is the engines' clocks as much as hers: the same on both engines; every shot's weight was the same on both. Her face shot is for a Sarge alone (measured on the Princess Croak hero). The guard's ram and her light arrows go through the same doors for a

hero as for a man (the code; not staged on this page).

The level a man first survives

For her harp shot and the hornet's sting. Her auto attack's law: a man on foot survives no shot at any rank; a heavy survives one from rank 0 (on MEGA from 70) and never two; Sarge's rows are the arithmetic. The ram and her face shot: every man but the rifleman survives the first from a full bar (the floor); a rifleman from the rank his HP passes the bite (never on EASY to HARD, never on MEGA). No man survives the jump-back crit. **Green** = he already does at the lab's EASY level 70; — = not even at rank 100.

			HER SHOTS IN A ROW · EASY–HARD (2 bar hits)				… MEGA (3)				HORNET STINGS		HER SHOTS TO KO · EASY–HARD, BY RANK						
man	HP / bar at 0	HP / bar at 100	1	2	3	4	1	2	3	4	1	2	r0	r10	r25	r50	r70	r90	r100
RIFLEMAN	3 / 3	45 / 8	—	—	—	—	—	—	—	—	0	2	1	1	1	1	1	1	1
M. GUNNER	8 / 3	65 / 8	—	—	—	—	—	—	—	—	0	0	1	1	1	1	1	1	1
BAZOOKA	16 / 3	142 / 8	0	—	—	—	70	—	—	—	0	0	2	2	2	2	2	2	2
MORTAR	3 / 3	40 / 8	—	—	—	—	—	—	—	—	0	2	1	1	1	1	1	1	1
MEDIC	2 / 2	40 / 7	—	—	—	—	—	—	—	—	0	8	1	1	1	1	1	1	1
FLAME	16 / 3	170 / 8	0	—	—	—	70	—	—	—	0	0	2	2	2	2	2	2	2
SWEEPER	3 / 3	42 / 8	—	—	—	—	—	—	—	—	0	2	1	1	1	1	1	1	1
RADIO MAN	3 / 3	42 / 8	—	—	—	—	—	—	—	—	0	2	1	1	1	1	1	1	1
GRENADIER	3 / 3	84 / 8	—	—	—	—	—	—	—	—	0	2	1	1	1	1	1	1	1
COMMANDO	16 / 4	152 / 9	0	—	—	—	70	—	—	—	0	0	2	2	2	2	2	2	2
SPY	2 / 2	50 / 7	—	—	—	—	—	—	—	—	0	8	1	1	1	1	1	1	1
SARGE · SABER	9 / 8	50 / 13	0	0	0	0	0	0	0	0	0	0	9	11	12	13	13	13	15
SARGE · PISTOL	8 / 4	50 / 12	0	0	0	0	0	0	5	10	0	0	5	7	8	12	12	12	13
SARGE · SHOTGUN	9 / 4	250 / 9	0	0	0	0	0	0	5	10	0	0	5	7	8	9	9	9	10
SARGE · SNIPER	6 / 4	40 / 9	0	0	0	0	0	0	5	10	0	0	5	7	8	9	9	9	10
SARGE · CLAYMORE	22 / 19	78 / 24	0	0	0	0	0	0	0	0	0	0	20	23	24	25	25	25	26

Read from the game's own HP and bar ladders at every rank 0–100 (the tenth-rank extras included; no gifts, soul or gear). Sarge's rows read the iron at the same rank. HER SHOTS TO KO by rank on EASY to HARD (2 bar hits a shot, landing at the damper's 50 %); MEGA's third bar hit KOs a Sarge in fewer (a level-90 pistol Sarge: the 8th shot, the 12th at level 70 on EASY); a man is the law.

The units

unit	attack	what a hit does	HP EASY	HP MED	HP HARD	HP MEGA	n EASY	n MED	n HARD	n MEGA	leak	tags
 **PRINCESS CROAK (the boss)** walks 24 px/s, keeps 240 px off your line and backs off at 96 px/s inside 160 px · lives 1 · 1 · 3 · 3	**HARP SHOT · TONGUE YOINK · FACE SHOT · LIGHT ARROWS** a harp shot at the nearest man within 400 px every 2.4 · 2 · 1.8 · 2 s · her tongue every 14 · 11 · 9 · 7 s at the nearest man within 380 px · a Sarge within 84 px: the face shot · the light arrows with the wave her half bar releases	her harp shot **kills**: any man on foot falls to the first that lands, a heavy (the bazooka, the flame and the commando, every machine) to the second (MEGA: below level 70, the first); Sarge and the heroes take **2 · 2 · 2 · 3 bar hits** (E · M · H · G) · the yoink's jump-back crit fells every man she licks · the face shot a bite of **49.5 HP** (MEGA 58.5) · the light arrows **push 220 px**	2706	3608	1804	7560	1	1	1	1	8	400
 **THE GNOME GUARD** walks 26 px/s · his shield turns every 2nd round from the front; bullets do half, blasts double · a fairy's bubble makes him BIG (×4 HP) · Princess Croak's summoned guards charge the line at ×2.2 once big	**RAM · JAB** braces 0.45 s then charges 380 px/s at a man ahead within 300 px, 100 px past him, then rests 1.6 s · jabs a man within 28 px every 1 s (0.2 s wind-up)	the ram a bite of **49.5 HP** (MEGA 58.5), landing whole: a rifleman falls, any other man stands his first from a full bar on 15 % of it (the saber turns it about half the time, THE RIOT SHIELD's 5 and the Claymore's counted block eat it; a jeep or the tank destroyed on the spear) · the jab **1 bar hit**	8	10	12	14	40	54	144 +3 a call	188 +4 a call	1	18
 **THE GNOME KNIGHT** MEDIUM up · walks 20 px/s · his shield turns EVERY round from the front (shoot him from behind or the flank); bullets do half, blasts double · a shield met by a light arrow from a Sarge's mod throws it back · the bazooka's rockets take ×2.4 off him, as off a machine (since v31.40)	**JAB** jabs a man within 28 px every 1.4 s (0.35 s wind-up); no ram	**2 bar hits**	—	18	21	25	—	4	8	10	2	36
 **THE GNOME SLINGER** walks 24 px/s	**GLOB** aims 0.6 s then slings at a man within 280 px, every 3 s and the aim (measured 3–3.5 · 3.1–3.5 · 3.1–3.6 · 3.1–3.5 s) · backs off from a man inside 120 px	**1 bar hit**	4	5	6	7	4	6	12	14	1	12

unit	attack	what a hit does	HP EASY	HP MED	HP HARD	HP MEGA	n EASY	n MED	n HARD	n MEGA	leak	tags
 **THE GNOME SAGE** MEDIUM up (one at the pond; HARD three; MEGA three at the open and two at the pond) · walks 22 px/s	**FIRE RINGS · FIREBALL · TOAD CURSE** the first fire at 6 s then every 12 s: the staff up 0.6 s, the rings 1.0 s, then 3 patches 56 px around their centres (112 px across) · the fireball the first after 4 s then every 9 s at the densest clump of your men within 420 px, flying 260 px/s with a ring closing under it (measured: thrown at 4 s, landed at 4.9 s) · the curse on a Sarge within 200 px, a 0.6 s cast, every 15 s	the fire **1 bar hit a beat**, a beat every 0.5 s for 5 s (10 beats) under each of your three rearmost men · the fireball **2 bar hits** on every man within 60 px of where it lands, and a fire patch there · the curse turns **Sarge** into a toad for 5 s (he hops at ×0.6, cannot fire or swing)	—	13	15	18	—	2	6	10	2	30
 **THE FAIRY** hovers 80 px behind the front man · Princess Croak's summoned fairies blow bubbles and never mend	**MEND · BUBBLE** the first bubble at 4 s, then every 10 s (0.6 s blow, 120 px/s); one big man a fairy at a time	no hit on your men: mends her court **1 HP a second** within 120 px and blows a bubble that makes the front gnome **BIG: ×4 HP, ×1.5 size, jabs ×1.5 as often**	3	4	5	5	2	4	4 +3 a call	4 +4 a call	1	14
THE BUBBLE a bubble gains on a man marching west; it stalls 12 px short of a man who stands still (measured)	**— (a buff in flight)** flies 120 px/s at the front gnome; BIG on arrival	none on your men; it pops on a green round (1 HP)	1	1	1	1	one a fairy every 10 s	one a fairy every 10 s	one a fairy every 10 s	one a fairy every 10 s	0	0
 **THE WASP** flies 78 px/s	**STING** dives 520 px/s at a man within 260 px, stings every man his pass crosses, climbs 2.2 s between dives (half that while a hornet is on the floor)	**1 bar hit** + the man is **HELD 2 s** (no march, no gun)	2	3	3	4	48	42	82	100	1	8

unit	attack	what a hit does	HP EASY	MED	HARD	MEGA	n EASY	MED	HARD	MEGA	leak	tags
 **THE HORNET** HARD and MEGA · flies 60 px/s · while he lives every wasp dives twice as often	**HEAVY STING** hangs 0.4 s over his man (the tell) then dives 420 px/s at a man within 300 px, climbs 2.6 s	**3 bar hits** + the man is **bowled over**	—	—	18	22	—	—	4	6	2	34
 **THE POND FROG** born from the pool's rim; never walks	**SPLAT · TONGUE** hops 460 px/s at a man within 150 px (one a hop, every 2 s) · the tongue at a man within 56 px every 2 s · beside a man: a tongue and a hop about every 3 s each (measured: 6 hits in 7 s, 2.76 bar hits landed)	the splat **1 bar hit** + a shove · the tongue **1 bar hit**	4	5	6	7	4	6	12	12	1	10

HP = each body as the palace's own wave rows bear it on that tier (measured in this read: the court ×1 · ×1.25 · ×1.5 · ×1.8, her own row ×16.9125 · ×22.55 · ×11.275 · ×47.25 over her 160 HP = the bar the designers' time-to-kill chart on EASY · MEDIUM · HARD and the player-damage-threshold pass on MEGA set; the dial ×2 · ×2.4 · ×6 · ×9 that multiplied her bar before v29.30 is still in the client and no longer sets it); a BIG guard has four times his. n = bodies named in the solo wave sheet (one commander), THE COURT RETURNS rows included; her summons come on top ("+N a call"). leak = lives it costs at the toy box. tags = dog tags paid when it falls. Our rounds on the gnomes: a guard's shield turns every 2nd round from the front and bullets do half (blasts double); a knight's shield turns every round from the front (measured: 5 of 5 rifle rounds, his bar unmoved). What a hit does is the blow as the game ships it: on your men it lands at 50 % (THE DAMPER, since v29.17), times the share; a KO is not dampened.

Princess Croak, move by move

move	EASY	MEDIUM	HARD	MEGA	how it works
HARP SHOT	**a man on foot:** **KO** · a heavy: 2 · every 2.4 s	**a man on foot:** **KO** · a heavy: 2 · every 2 s	**a man on foot:** **KO** · a heavy: 2 · every 1.8 s	**a man on foot:** **KO** · a heavy: 2 · every 2 s (below level 70: KO)	HER AUTO ATTACK KILLS (since v32.27): the nearest man within 400 px (a jetting, cloaked or dragged man is not picked); a shot that LANDS takes at least his whole bar from a man on foot, so he falls to the first whatever his level, and half a heavy's (the bazooka, the flame and the commando, every machine), so he falls to the second; on MEGA a heavy below level 70 falls to the first. Measured on all eleven men on every tier on both engines (one shot each; the bazooka, the flame and the commando two) and by level (a level-100 rifleman and a level-1 medic: one; a level-0 flame and a level-100 commando: two; on MEGA a level-69 bazooka one, a level-70 two). Sarge and the heroes take her shot as before: 2 · 2 · 2 · 3 bar hits (E · M · H · G) at the damper's 50 % (a level-70 pistol Sarge falls to the 12th). Every save keeps its say: a shot the shield, the dome or a dodge turns takes nothing. Measured every 2.42 · 2.02 · 1.82 · 2.02 s. She glides to 240 px from your line and backs off east at 96 px/s when a man closes inside 160 px.
TONGUE YOINK · JUMP-BACK CRIT	every **14 s** · **KO**	every **11 s** · **KO**	every **9 s** · **KO**	every **7 s** · **KO**	Her tongue (since v31.45) and its crit (since v32.27): every 14 · 11 · 9 · 7 s from tell to tell (the first after half of it; 5 s while she stands on her stage after a retreat) she picks the nearest man within 380 px with a clear line (rocks, the tall columns and the pool block it; never Sarge, a hero, a machine, a rider or a gun; never a man within 70 px), a 0.6 s tell (her cheeks swell), the tongue out, the reel (0.8 s) to her feet and a 1 s daze; then her hop back (80 px, 0.3 s) and one shot that ALWAYS crits (×2 her auto attack): every man she can lick falls, a heavy too. Measured on a rifleman, a bazooka and a commando on every tier, both engines: down 0.31–0.32 s after the landing; a pistol Sarge in her reach was never licked. 5 % of her bar landed during the reel snaps the tongue; with no man in reach a Dogfighter is pulled down and shot down.

move	EASY	MEDIUM	HARD	MEGA	how it works
FACE SHOT	**49.5 HP** · 0.47 s	**49.5 HP** · 0.47 s	**49.5 HP** · 0.47 s	**58.5 HP** · 0.47 s	A SARGE within 84 px (the Claymore, the pistol, any iron; never a man or a hero: measured). The 0.12 s crouch tell, the 80 px hop back (0.3 s), and the shot in the face. NO INSTANT KO (since v29.70): it takes 49.5 HP (MEGA 58.5) off his sheet, landing whole, and no single hit fells him from a full bar (since v31.00: he stands on 15 % of it, the next one fells him). The saber turns it about half the time now, THE RIOT SHIELD and the Claymore's counted block eat it whole, the deflect dome turns it. Measured: a pistol Sarge at 60 px left on 1.65 of his 11 bar hits; kept 60 px off her, the shotgun Sarge fell to face shot 9 · 9 · 10 · 9 (the plate's 5 first) and the Claymore to 3 · 3 · 3 · 3 (E · M · H · G); a rifleman at 60 px took her harp instead.
LIGHT ARROWS	**a push of 220 px** · at 50 %	**a push of 220 px** · at 50 %	**a push of 220 px** · at 50 %	**a push of 220 px** · at 50 % and 25 %	With the wave her half bar releases (every tier: EASY's since v31.40) she roots and charges 2 s, then three bolts fly west from her, 120 px apart, 900 px/s. THE PUSH (since v31.40): the light kills nobody; every man whose figure a bolt's line crosses is pushed 220 px west (measured 200–208 px on foot), a machine 0.3 of it and never into the water (measured: the jeep slid 13–51 px); a column or a fountain stops a pushed man with a bump (one bar hit), the pool or the water stairs off a plank are a fall. Measured: the three men on her three lines pushed on every tier, a medic 120 px under the lowest line untouched (60 and 80 px were not clear: the line crosses his figure). MEGA fires it again at a quarter ("Punishment" in E Minor); on HARD and MEGA the arrow act also comes with her retreats.
THE CATCH	she takes **676.5 HP** (25 % of her bar)	she takes **902 HP** (25 % of her bar)	she takes **451 HP** (25 % of her bar)	she takes **1,890 HP** (25 % of her bar)	A charged commando standing on a line (or covering a man with THE BUDDY SHIELD), the blue wall, the teal's save or a swinging Claymore CATCHES a bolt: it flies back east at her, every gnome it crosses falls, and she takes a quarter of her bar (measured on every tier: the gnome in its path fell, her bar –25 %). A commando catches ONE bolt a fight (measured on MEGA: the second volley pushed him).
THE GATES	75 %: GUARDS! TO ME! 50 %: THE GUARD TURNS OUT 30 %: THE POND! EVERYONE OUT OF THE POND! then THE COURT RETURNS ×5 at 25 · 20 · 15 · 10 · 5 % (2 gnome, 4 wasp)	75 %: GUARDS! TO ME! 50 %: LET ME SHOW YOU TRUE POWER. — THE LIGHT ARROWS 30 %: THE POND! EVERYONE OUT OF THE POND! then THE COURT RETURNS ×5 at 25 · 20 · 15 · 10 · 5 % (3 gnome, 3 wasp)	75 %: GUARDS! TO ME! 50 %: LET ME SHOW YOU TRUE POWER. — THE LIGHT ARROWS 30 %: THE POND! EVERYONE OUT OF THE POND! then THE COURT RETURNS ×8 at 27 · 24 · 21 · 18 · 15 · 12 · 8 · 4 % (6 gnome, 4 wasp)	75 %: GUARDS! TO ME! 50 %: LET ME SHOW YOU TRUE POWER. — THE LIGHT ARROWS 30 %: THE POND! EVERYONE OUT OF THE POND! 25 %: AGAIN. — THE LIGHT ARROWS then THE COURT RETURNS ×8 at 22 · 20 · 17 · 15 · 12 · 9 · 7 · 4 % (8 gnome, 5 wasp)	Her court comes on her bar (since v31.39): a wave HOLDS while her court stands, then comes when her bar falls to its mark, or after the idle clock (56 · 48 · 42 · 42 s with her court down and nobody denting her); THE COURT RETURNS come on her bar below the last gate; a wave cleared pays a purse of 120 · 150 · 180 · 200 dog tags. The gates latch once.
HER LIVES	**1** bar	**1** bar	**3** bars of 1,804	**3** bars of 7,560	EASY and MEDIUM: her first zero is her fall. HARD and MEGA: THE FALL THAT ISN'T — at zero with bars left she flies back to her stage (≤ 1 s), holds 0.8 s, rises with a whole new bar and acts: HARD "summon" then "arrow", MEGA "both" twice. Measured on HARD: the retreat at 0.9 s, the court at 2.2 s; the second zero brought the arrows. Her third zero is her fall.
THE BUBBLE COURT	—	—	**6 guards + 6 fairies**	**8 guards + 8 fairies**	"MY FAIRIES! Get big my darlings!": the guards appear at her stage at ×1.5 / ×1.8 HP, each fairy on her own guard, twice the court since v32.27 (HARD 3 + 3 before, MEGA 4 + 4); the bubbles make them BIG (×4 HP) and the big guards CHARGE your line at ×2.2 speed (measured: 6 guards and 6 fairies on HARD, 6 big within 4 s). Her fairies never mend.
THE BAR CLOCK	—	—	every **60 s**	every **45 s**	A later bar that stands 60 s (HARD) / 45 s (MEGA) calls THE BUBBLE COURT again (measured: the call at 60 s and 45 s). The fight always moves.
THE WAVE AFTER HER FIRST BAR	—	—	+6 guards + 6 fairies	+8 guards + 6 fairies	Once her first bar is down every COURT RETURNS wave also fields charging guards with a fairy each, twice as many since v32.27 (the code's own rows; not staged on this page).

move	EASY	MEDIUM	HARD	MEGA	how it works
TWICE THE COURT	**98** bodies (49)	**118** bodies (59)	**272** bodies (136)	**344** bodies (172)	Since v32.27 every body on her side comes twice, every tier, solo and party: the opening court, every wave and every return, the pond frogs, the bubble court, the charging guards, the fountain and teleport arrivals. One of her, always; her bar, her court's bars and hits as before. Measured: the opening court as the lab fields it, every kind twice the sheet's old row on every tier (EASY 8 guards · 4 slingers · 2 fairies).
WALK	24 px/s	24 px/s	24 px/s	24 px/s	Her glide; 96 px/s on the back-off. A press she cannot gain a pixel on for 1.2 s is given up for a wander step (the throne room's columns and fountains).
HER BAR (one commander)	**2,706**	**3,608**	**1,804**	**7,560**	Set by the designers' time-to-kill chart on EASY · MEDIUM · HARD and the player-damage-threshold pass on MEGA; the game writes it on her row as 160 HP × 16.9125 · 22.55 · 11.275 · 47.25 (the dial ×2 · ×2.4 · ×6 · ×9 that multiplied her bar before v29.30 is still in the client and no longer sets it). Over her lives (1 · 1 · 3 · 3) that is 2,706 · 3,608 · 5,412 · 22,680 in all. Her fall ends the fight. With more commanders the bar grows (the table below: solo × (1 + 0.9 a commander past the first), rounded to the HP).
THE TOY BOX	−8 lives if she reaches it	−8 lives if she reaches it	−8 lives if she reaches it	−8 lives if she reaches it	Every other unit costs its LEAK (1–2) of the box's lives.

Her bar by the number of commanders

	1 commander	2 commanders	3 commanders	4 commanders	5 commanders	10 commanders
EASY	2,706	5,141	7,577	10,012	12,448	24,625
MEDIUM	3,608	6,855	10,102	13,350	16,597	32,833
HARD	1,804	3,428	5,051	6,675	8,298	16,416
MEGA	7,560	14,364	21,168	27,972	34,776	68,796

Read from the game's own wave sheet with 1, 2, 3, 4, 5 and 10 commanders in the fight. HARD and MEGA: three such bars (her lives: 5,412 / 22,680 in all with one commander). Each party cell is the solo bar × (1 + 0.9 a commander past the first), rounded to the HP. One Princess Croak at every seat count; her court scales with the party.

What changes from EASY to MEGA

	EASY	MEDIUM	HARD	MEGA
Her harp shot on a man on foot · a heavy · Sarge (bar hits) · seconds between	KO · 2 · 2 · 2.4	KO · 2 · 2 · 2	KO · 2 · 2 · 1.8	KO · 2 (KO below 70) · 3 · 2
Her tongue (seconds between yoinks; the jump-back crit fells)	14 s	11 s	9 s	7 s
No instant KO: the ram and her face shot take (HP, landing whole)	49.5	49.5	49.5	58.5
One blow reaches at most (men)	3	4	4	5
An enemy's blow on your men lands at (THE DAMPER)			50 %	
Her light arrows (a push of 220 px)	at 50 %	at 50 %	at 50 %	**at 50 % and 25 %**
Her lives (bars)	1	1	3	3
Her bar (one commander)	2,706	3,608	1,804	7,560
Who sets her bar	the designers' time-to-kill chart	the designers' time-to-kill chart	the designers' time-to-kill chart	the player-damage-threshold pass

	EASY	MEDIUM	HARD	MEGA
Her bars in all (lives × bar, one commander)	2,706	3,608	5,412	22,680
The court's HP	×1	×1.25	×1.5	×1.8
Guards · slingers · knights · sages in the sheet	40 · 4 · — · —	54 · 6 · 4 · 2	144 · 12 · 8 · 6	188 · 14 · 10 · 10
Wasps · hornets · frogs · fairies in the sheet	48 · — · 4 · 2	42 · — · 6 · 4	82 · 4 · 12 · 4	100 · 6 · 12 · 4
Bodies in the sheet, solo (she is not counted; twice the court since v32.27)	98 (49 before)	118 (59 before)	272 (136 before)	344 (172 before)
The idle clock (a wave comes anyway after, her court down)	56 s	48 s	42 s	42 s
A wave cleared pays (dog tags)	120	150	180	200
THE COURT RETURNS (waves, on her bar)	5 × at 25 · 20 · 15 · 10 · 5 %	5 × at 25 · 20 · 15 · 10 · 5 %	8 × at 27 · 24 · 21 · 18 · 15 · 12 · 8 · 4 %	8 × at 22 · 20 · 17 · 15 · 12 · 9 · 7 · 4 %
THE BUBBLE COURT at a retreat (guards + fairies) · the bar clock	—	—	6 + 6 · 60 s	8 + 8 · 45 s
The room	day	day	day	**moonlit** (night)
The lab's stock army level	70	77	83	90

The wave sheets (solo, one commander)

EASY · 98 bodies · level 70

1. **PRINCESS CROAK TAKES THE STAIR** (on the clock)
   8 gnome guard, 4 slinger, 2 fairy · PRINCESS CROAK (bar 2,706)
2. **GUARDS! TO ME!** (waits for her court to fall, then princess at 75% (or 56 s with the court down))
   4 wasp, 2 pond frog
3. **THE GUARD TURNS OUT** (waits for her court to fall, then princess at 50% (or 56 s with the court down))
   6 gnome guard, 4 wasp
4. **THE POND! EVERYONE OUT OF THE POND!** (waits for her court to fall, then princess at 30% (or 56 s with the court down))
   2 pond frog, 6 gnome guard
5. **GUARDS! TO ME!** (waits for her court to fall, then princess at 25% (or 56 s with the court down))
   4 gnome guard, 8 wasp
6. **GUARDS! TO ME!** (waits for her court to fall, then princess at 20% (or 56 s with the court down))
   4 gnome guard, 8 wasp
7. **GUARDS! TO ME!** (waits for her court to fall, then princess at 15% (or 56 s with the court down))
   4 gnome guard, 8 wasp

MEDIUM · 118 bodies · level 77

1. **PRINCESS CROAK TAKES THE STAIR** (on the clock)
   10 gnome guard, 6 slinger, 2 fairy · PRINCESS CROAK (bar 3,608)
2. **GUARDS! TO ME!** (waits for her court to fall, then princess at 75% (or 48 s with the court down))
   4 knight, 2 fairy, 8 wasp, 4 pond frog
3. **LET ME SHOW YOU TRUE POWER.** (waits for her court to fall, then princess at 50% (or 48 s with the court down))
   8 gnome guard, 4 wasp
4. **THE POND! EVERYONE OUT OF THE POND!** (waits for her court to fall, then princess at 30% (or 48 s with the court down))
   2 pond frog, 6 gnome guard, 2 sage
5. **GUARDS! TO ME!** (waits for her court to fall, then princess at 25% (or 48 s with the court down))
   6 gnome guard, 6 wasp
6. **GUARDS! TO ME!** (waits for her court to fall, then princess at 20% (or 48 s with the court down))
   6 gnome guard, 6 wasp
7. **GUARDS! TO ME!** (waits for her court to fall, then princess at 15% (or 48 s with the court down))
   6 gnome guard, 6 wasp

HARD · 272 bodies · level 83

1. **PRINCESS CROAK TAKES THE STAIR** (on the clock)
   20 gnome guard, 12 slinger, 2 fairy · PRINCESS CROAK (bar 1,804)
2. **GUARDS! TO ME!** (waits for her court to fall, then princess at 75% (or 42 s with the court down))
   8 knight, 2 fairy, 14 wasp, 4 hornet, 6 pond frog
3. **LET ME SHOW YOU TRUE POWER.** (waits for her court to fall, then princess at 50% (or 42 s with the court down))
   16 gnome guard, 4 wasp
4. **THE POND! EVERYONE OUT OF THE POND!** (waits for her court to fall, then princess at 30% (or 42 s with the court down))
   6 pond frog, 12 gnome guard, 6 sage
5. **GUARDS! TO ME!** (waits for her court to fall, then princess at 27% (or 42 s with the court down))
   12 gnome guard, 8 wasp
6. **GUARDS! TO ME!** (waits for her court to fall, then princess at 24% (or 42 s with the court down))
   12 gnome guard, 8 wasp
7. **GUARDS! TO ME!** (waits for her court to fall, then princess at 21% (or 42 s with the court down))
   12 gnome guard, 8 wasp

MEGA · 344 bodies · level 90

1. **PRINCESS CROAK TAKES THE STAIR** (on the clock)
   24 gnome guard, 14 slinger, 2 fairy, 6 sage · PRINCESS CROAK (bar 7,560)
2. **GUARDS! TO ME!** (waits for her court to fall, then princess at 75% (or 42 s with the court down))
   10 knight, 2 fairy, 16 wasp, 6 hornet, 6 pond frog
3. **LET ME SHOW YOU TRUE POWER.** (waits for her court to fall, then princess at 50% (or 42 s with the court down))
   20 gnome guard, 4 wasp
4. **THE POND! EVERYONE OUT OF THE POND!** (waits for her court to fall, then princess at 30% (or 42 s with the court down))
   6 pond frog, 16 gnome guard, 4 sage
5. **AGAIN.** (waits for her court to fall, then princess at 25% (or 42 s with the court down))
   nothing new: her second volley
6. **GUARDS! TO ME!** (waits for her court to fall, then princess at 22% (or 42 s with the court down))
   16 gnome guard, 10 wasp
7. **GUARDS! TO ME!** (waits for her court to fall, then princess at 20% (or 42 s with the court down))
   16 gnome guard, 10 wasp

8. **GUARDS! TO ME!** (waits for her court to fall, then princess at 10% (or 56 s with the court down))
   4 gnome guard, 8 wasp
9. **GUARDS! TO ME!** (waits for her court to fall, then princess at 5% (or 56 s with the court down))
   4 gnome guard, 8 wasp

8. **GUARDS! TO ME!** (waits for her court to fall, then princess at 10% (or 48 s with the court down))
   6 gnome guard, 6 wasp
9. **GUARDS! TO ME!** (waits for her court to fall, then princess at 5% (or 48 s with the court down))
   6 gnome guard, 6 wasp

8. **GUARDS! TO ME!** (waits for her court to fall, then princess at 18% (or 42 s with the court down))
   12 gnome guard, 8 wasp
9. **GUARDS! TO ME!** (waits for her court to fall, then princess at 15% (or 42 s with the court down))
   12 gnome guard, 8 wasp
10. **GUARDS! TO ME!** (waits for her court to fall, then princess at 12% (or 42 s with the court down))
    12 gnome guard, 8 wasp
11. **GUARDS! TO ME!** (waits for her court to fall, then princess at 8% (or 42 s with the court down))
    12 gnome guard, 8 wasp
12. **GUARDS! TO ME!** (waits for her court to fall, then princess at 4% (or 42 s with the court down))
    12 gnome guard, 8 wasp

8. **GUARDS! TO ME!** (waits for her court to fall, then princess at 17% (or 42 s with the court down))
   16 gnome guard, 10 wasp
9. **GUARDS! TO ME!** (waits for her court to fall, then princess at 15% (or 42 s with the court down))
   16 gnome guard, 10 wasp
10. **GUARDS! TO ME!** (waits for her court to fall, then princess at 12% (or 42 s with the court down))
    16 gnome guard, 10 wasp
11. **GUARDS! TO ME!** (waits for her court to fall, then princess at 9% (or 42 s with the court down))
    16 gnome guard, 10 wasp
12. **GUARDS! TO ME!** (waits for her court to fall, then princess at 7% (or 42 s with the court down))
    16 gnome guard, 10 wasp
13. **GUARDS! TO ME!** (waits for her court to fall, then princess at 4% (or 42 s with the court down))
    16 gnome guard, 10 wasp

She opens on her top step; her court walks on behind her over the first seconds. A wave holds while her court stands, then comes when her bar falls to its mark, or after the idle clock with her court down (since v31.39); THE COURT RETURNS come on her bar too. Every count here is twice the old row (since v32.27): the sheet as the lab fields it.

How this was read and checked

The live client v32.36 (index.html md5 055c105b750dc1652be801b7b4e14f9f, the Big Mac's main at the time of this read, which carries CARD-PCH) was opened headless on Maker 1 and read with the game's own functions, THE DAMPER's table by name among them: every table, every tier's wave sheet, a real lab fight per tier for the stock army and its level, every man's HP and bar at every rank, her bar for 1 to 10 commanders. Then each hit was **measured on the real machines** in a BOSS LAB fight on every tier, every attacker born through the palace's own wave row so the tier's stamps ride: THE DAMPER in every lab fight (the game's own share on each tier equals its cell, and every hit on our men rode it); her harp shot on all eleven men and Sarge's five irons, each to his KO, and on the six heroes for 60 s; her auto attack by level (a heavy at 69 and 70, a flame at 0, a commando and a rifleman at 100, a medic at 1); her tongue's jump-back crit on a rifleman, a bazooka and a commando (and that a Sarge in reach is never licked); her face shot on a saber and a pistol Sarge (and that a rifleman, a hero and a Sarge at 100 px are spared); her light arrows with the wave her half bar releases, on three men on her three lines, a jeep in the middle and a medic clear of them (the push), the commando's catch and the ray thrown back at her and her gnome, MEGA's second volley; the opening court as the lab fields it (twice the sheet's old rows); the sage's fireball on a clump; the guard's jab and ram (and the scout jeep, the V2 jeep and the tank on the spear), the knight's jab and shield, the guard's shield and armour under our rifle, the slinger's glob, the sage's toad curse and fire, the fairy's mend and bubble, the wasp's sting and the hold, the hornet's sting and the bowl-over, the frog's splat and tongue; on HARD and MEGA her retreat, THE BUBBLE COURT, the big guards' charge, the bar clock and the second zero; every body as the tier bears it. Chrome 309/309 · WebKit 309/309 checks passed, zero page errors. The two engines' READS are identical (checked), and every damage fact in the two proofs agrees (1560 facts checked across the engines: each hit's weight, every KO and every save, every body). What differs between Chrome and WebKit is dice and clocks, and this page prints both where it matters: the saber Sarge's harp shots landed to his KO (34/18 · 59/66 · 16/29 · 12/26 on E · M · H · G, after 147/67 · 252/274 · 47/136 · 40/141 s); the gnome's ram on 16 fresh saber Sarges (0/0 · 0/0 · 0/0 · 0/0 KO'd); the frame a hit lands on, and the slinger's wander.

Sources (the code's own names): FDEF (princess · gnome · gslinger · gsage · gknight · fairy · fbubble · wasp · hornet · pfrog); updateFoePrincess (PC_SHOT, the halt and back-off, pcNearest, pcFaceShot → knock(rolled), the gates → pcFire), the 'lightarrow' shot branch (pkbArrowHit → ggShove, PC_ARROW_PUSH, PKB_VEH; cmdoCharged/cmdoCovers, tkDeflT, mblRay, arrowBack → hurt under TK_REFL), pflHold / pflRelease / pflArrowTry (PFL_IDLE, PFL_RET_K, PFL_PURSE), PHX_* (phxFloor on tanHit, phxJbStart / phxJbShot, spawnGroup x PHX_COURT), pykTell / pykLand (PYK_CD, PYK_REACH), knockKill → bdc2Price (BDC2_K) → tanHit, rzbFloorStand (RZB_FLOOR, RZB_CAP), pkbFbTick / pkbFbLand (PKB_FB), PC_ARROW_TIER, PC_LIVES, pcRetreat / pcRetTick (PC_RET, PC_RET_ACTS), pcSummon / pcSummonIn (PC_SUMMON, PC_COURT_HPX), PC_BAR_CLK, pcwWave (PCW), PC_GIANT_SPDX; updateFoeTrike (the ram bag → knock; the vehicle branch → knockKill), gnJabTick, gnSlingTick + tanShoot (def.pips → pipN), gnCasterTick (fsIgnite → fsTick: tanHit fire beats; toadT), gnFairyTick / gnBubbleTick (big, bigX), updateFoeSwoop (sting.pips → tanHit, kyHeldT, ggShove), updateFoePounce (pounce.pips, tongue), hurt (the gnomes' shield shN, gnArmorX); tanHit, DMG_DAMP / dmpShare / dmpX (THE DAMPER inside tanHit's pip write), DSM (k 0.9), dsmH / dsmX, dhtH, gritCap, radioTick (RADIO_DROP_T, held in the harp staging), mgCtr, enhSoakM, saberBlocks, sargeAbsorbs (ENH_SHIELD_N), cmdAbsorbs; raidCfg's palace branch (the court's cells, hpGate rows with clk, PC_GATE_CLK, PC_RETURN_CLK; PC_HPX stays in the client and no longer sets her row: rxtApply / RXT.live and pdtScale / PDT.bars write her row's hpX over her base HP), blbSheet('palace', tier, N), blbStockDef (BLB_STOCK_LV.palace), blbLaunch; bfHpAt, usBfHP, sgBfHP, heroBfHP. Kit: design/boss-princess-unit-breakdown-1031/ (kit/readPalace.mjs · kit/provePalace.mjs · kit/build.py · kit/render.mjs · kit/export_dash.py; rig/lrbShared.mjs; data/; proof/logs/). Zero game bytes.